Senior Data Engineer

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PROFESSIONAL SUMMARY

Professional Summary:

- **10 plus years of IT background** in end-to-end data analytics solutions.
- **Azure DevOps Expertise:** Proficiently build reusable YAML pipelines in Azure DevOps, create CI/CD pipelines using cloud-native architectures on Azure Cloud, and implement Git flow branching strategy.
- **Cloud Native Technologies:** Solid grasp of Azure services including Databricks, Data Factory, Data Lake, and Function Apps. Proficient with SQL and experienced in Agile, SAFe, and DevOps methodologies. Developed Spark streaming modules for RabbitMQ and Kafka data ingestion.
- **DevOps and Scripting Proficiency:** Skilled in PowerShell scripting, Bash, YAML, JSON, GIT, Rest API, and Azure Resource Management (ARM) templates. Implement CI/CD standards, integrate security scanning tools, and manage pipelines effectively.
- **Windows Scripting and Cloud Containerization**: Proficient in scripting and debugging within Windows environments. Familiarity with container orchestration, Kubernetes, Docker, and AKS.
- **Efficient Data Integration:** Expertise in designing and deploying SSIS packages for data extraction, transformation, and loading into Azure SQL Database and Data Lake Storage. Configure SSIS Integration Runtime for Azure execution and optimize package performance.
- **Data Visualization and Analysis:** Create data visualizations using Python, Scala, and Tableau. Develop Spark scripts with custom RDDs in Scala for data transformation and actions. Conduct statistical analysis on healthcare data using Python and various tools.
- **Azure Data Services:** ETL expertise using Azure Data Factory, T-SQL, Spark SQL, and U-SQL Azure Data Lake Analytics. Ingest data to Azure Services and process within Azure Databricks.
- **Hadoop Proficiency:** Strong support experience across major Hadoop distributions - Cloudera, Amazon EMR, Azure HDInsight, Hortonworks. Proficient with Hadoop tools - HDFS, MapReduce, Yarn, Spark, Kafka, Hive, Impala, HBase, Sqoop, Airflow, and more.
- **Azure Cloud and Big Data Tools:** Working knowledge of Azure components - HDInsight, Databricks, Data Lake, Blob Storage, Data Factory, SQL DB, SQL DWH, Cosmos DB. Hands-on experience with Spark using Scala and PySpark.
- **Database Migration:** Expertise in migrating SQL databases to Azure Data Lake, Azure SQL Database, Data Bricks, and Azure SQL Data Warehouse. Proficient in access control and migration using Azure Data Factory.
- **Cloud Computing and Big Data Tools:** Proficient in Azure Cloud and Big Data tools - Hadoop, HDFS, MapReduce, Hive, HBase, Spark, Azure Cloud, Kafka, Flume, Avro, Sqoop, PySpark.
- **Real-time Data Solutions:** Build real-time data pipelines and analytics using Azure components like Data Factory, HDInsight, Azure ML Studio, Stream Analytics, Azure Blob Storage, and Microsoft SQL DB.
- **Database Expertise:** Work with SQL Server and MySQL databases. Skilled in working with Parquet files, parsing, and validating JSON formats. Hands-on experience in setting up workflows with Apache Airflow and Oozie.
- **API Development and Integration:** Develop highly scalable and resilient RESTful APIs, ETL solutions, and third-party platform integrations as part of an Enterprise Site platform.
- **IDE and Version Control**: Proficient use of IDEs like PyCharm, IntelliJ, and version control systems SVN and Git.

TECHNICAL SKILLS:

Big Data Technologies	Kafka, Cassandra, Apache Spark, HBase, Impala, HDFS, MapReduce, Hive, Dataproc ,Pig, Sqoop, Flume, Oozie, Zookeeper, AKS, Cosmos DB, Strom
Hadoop Distribution	Cloudera CDH, Horton Works HDP

Programming Languages	SQL, Python, Java, PySpark, Scala, Shell Scripting, Regular Expressions
Spark components	RDD, Spark SQL (Data Frames and Datasets), Spark Streaming, Druid, Apache, Presto, ozone
Cloud Infrastructure	Azure, AWS, GCP
Databases	Oracle, Teradata, MySQL, SQL Server, NoSQL Databases (HBase, MongoDB)
Version Control	Git
Build Tools	Maven, SBT
Containerization Tools	Kubernetes, Docker
Reporting Tools	Power BI, Tableau

EDUCATION:

I obtained my Bachelor's degree in **ECE** from **JNTUH** in **2014**.

Professional Experience

Client: Optum, MN Jan 2024- Present
Title: Senior Data Engineer
Roles and Responsibilities:

- Worked on **Apache Spark** data processing project to process data from **RDBMS** and several data streaming sources and developed **Spark** applications using **Python** on **AWS EMR.**
- Designed and deployed multi-tier applications leveraging **AWS** services like (EC2, Route 53, S3, RDS, DynamoDB) focusing on high availability, fault tolerance, and **auto-scaling** in AWS **Cloud Formation**.
- Configured and launched **AWS EC2** instances to execute Spark jobs on **AWS Elastic Map Reduce (EMR).**
- Performed data transformations using Spark **Data Frames**, **Spark SQL**, **Spark File formats**, **Spark RDDs**.
- Implemented efficient data extraction from various sources such as relational databases, CSV files, and JSON documents using Spark's connectors and Snowflake's native integration capabilities and transformed data from different files (Text, CSV, JSON) using **Python** scripts in **Spark**.

- Designed and optimized **Spark** jobs to perform data cleansing, transformation, and aggregation tasks on large datasets before loading them into **Snowflake**. Worked on SQL queries with Presto.
- Developed and maintained Java-based applications, collaborating with cross-functional teams to meet project deadlines.
- Optimized data models, implemented effective partitioning strategies, and fine-tuned queries for enhanced scalability and reduced latency.
- Collaborated with data scientists to integrate Cosmos DB with Azure Synapse Analytics, enabling seamless data analytics and reporting.
- Integrated **Snowflake's SnowSQL CLI** and **Python SDK** with Spark to enable programmatic interactions with Snowflake, including creating databases, tables, and managing user permissions. Worked with Dataproc with spark.
- Led the design and deployment of containerized applications on Azure Kubernetes Service(AKS).
- Loaded data from various sources like **RDBMS** (MySQL, Teradata) using **Sqoop** jobs**.** Working with **Java J2EE.**
- Handled **JSON** datasets by writing custom **Python** functions to parse through JSON data using **Spark.**
- Implemented security best practices for AKS, including Azure Security Center configurations and Role-Based Access Control (RBAC).
- Developed a preprocessing job using **Spark Data Frames** to flatten **JSON** documents to **flat files**.
- Performed wide, narrow transformations, actions like filter, Lookup, Join, count, etc. on **Spark Data Frames**.
- Worked with Parquet files and Impala using **PySpark,** and Spark Streaming with RDDs and **Data Frames**.
- Utilized **Spark** to efficiently ingest and process large volumes of data from diverse sources such as databases, files, and streaming platforms into **Power BI.** Experience with kudu**.** Worked with the administration.
- Designed and built optimized data models using **Spark** for efficient querying and aggregation in **Power BI**. Leveraged Spark's advanced features like partitioning and columnar storage for improved performance.
- Aggregated logs data from various servers and made them available in downstream systems for analytics by using **Apache Kafka**. Improved **Kafka** performance and implemented security. Worked with Apache ozone.
- Developed batch and streaming processing apps using **Spark API**s for functional pipeline requirements.
- Used cloud shell SDK in GCP to configure the services Data Proc, Storage, BigQuery.
- Developed and deployed machine learning models for predictive analytics, recommendation systems, and natural language processing, utilizing Python and popular ML libraries.
- Automated data storage from streaming sources to AWS data lakes like **S3**, **Redshift** and **RDS** by configuring **AWS Kinesis (Data Firehose)**.
- Worked with object storage with AWS S3.
- Managed resources and scheduling across the cluster using **Azure Kubernetes Service.**
- Cleaned and handled missing values in data using **Python** by backward-forward filling methods and applied Feature engineering, normalize and label encoding techniques using **Python Scikit-learn** preprocessing.
- Stored data into various tiers of **AWS S3** based on business requirements and frequency of data access.
- Performed reporting analytics on data from AWS stack by connecting it to BI tools (**Tableau**, **Power BI**).
- Worked with database administrating team on **SQL optimization** for databases like **Oracle**, **MySQL**, **MS SQL**.
- Created visually appealing and interactive reports, dashboards, and visualizations in **Power BI** using Spark-processed data. Worked with Minio.
- Established seamless integration between **Spark** data processing and **Power BI** to enable real-time or near real-time data updates. Utilized **Power BI** connectors and APIs to connect Spark-processed data to Power BI reports and dashboards. Processed large volumes of high-velocity data using Strom. Work experience with Java.
- Assisted in configuring and implementing **DynamoDB** cluster nodes on **AWS EC2** instances.
- Identified executor failures, data skewness, and runtime issues by monitoring **Spark** apps through **Spark UI**.
- Automated deployments and routine tasks using **UNIX Shell Scripting**.
- Collaborated with the Data Science team building machine learning models on **Spark EMR** cluster to deliver the data needs under business requirements. Worked with analysis using Druid.
- Worked in an agile environment to implement projects and enhancements with weekly SCRUMs.

Environment: Hadoop 2.x, Spark v2.0.2, Hive, Sqoop, Kafka, Spark streaming, ETL, Cosmos DB, GCP, Scala, Python (Pandas, NumPy), PySpark, GIT (version control), Snowflake, Power BI, Java, MySQL, MS SQL, AKS, MongoDB, AWS (EC2, S3, EMR, RDS, Lambda, Kinesis, Redshift, Cloud Formation)

Client: Advent Health, FL Dec 2021 – Dec 2023
Title: Data Engineer
<u>Roles and Responsibilities:</u>

- Designed and deployed data pipelines using Azure cloud platform **(HDInsight, DataLake, DataBricks, Blob Storage, Data Factory, Synapse, SQL, SQL DB, DWH, and Data Storage Explorer)**.
- Developed custom-built **ETL** solution, batch processing, and real-time data ingestion pipeline to move data in and out of the Hadoop cluster using **PySpark** and **Shell Scripting**.
- Integrated on-premises data **(MySQL, Cassandra)** with cloud **(Blob Storage, Azure SQL DB)** and applied transformations to load back to **Azure Synapse** using **Azure Data Factory**.
- Built and published Docker container images using **Azure Container Registry** and deployed them into **Azure Kubernetes Service (AKS)**.
- Imported metadata into **Hive** and migrated existing tables and applications to work on **Hive** and **Azure.**
- Created complex data transformations and manipulations using **ADF** and **Scala**. Experience with Worked with Dataproc with Hadoop.
- Conducted code reviews with Java and implemented best practices to ensure code quality and maintainability.
- Collaborated with cross-functional teams to gather and understand business requirements, translating them into data and ML solutions. Experience with AKS.
- Configured **Azure Data Factory (ADF)** to ingest data from different sources like relational and non-relational databases to meet business functional requirements. Experience with Minio.
- Developed and optimized ETL pipelines using **Apache Spark** on **Azure Databricks** to extract, transform, and load data from various sources into data lakes and warehouses.
- Utilized **PySpark** to perform complex data transformations, data cleansing, and data enrichment tasks within **Azure Databricks** notebooks.
- Implemented best practices for AKS cluster provisioning, scaling, and maintenance.
- Automated and orchestrated data workflows using tools like Apache Airflow, ensuring data pipelines run efficiently.
- Implemented monitoring solutions and conducted performance analysis to identify and resolve issues in **Azure Databricks** clusters and jobs, ensuring system reliability and stability. Experience working with Strom for large data.
- Developed a comprehensive data quality framework for the BI environment, defining standards, rules, and best practices for data cleanliness, accuracy, completeness, and consistency. Big data analysis using Druid.
- Implemented monitoring solutions using **Azure Monitor** to track the performance of data pipelines and **Power BI** reports. Experience with Apache ozone. SQL operations on Presto.
- Optimized workflows by building DAGs in **Apache Airflow** to schedule the ETL jobs and implemented additional components in Apache **Airflow** like Pool, Executors, and multi-node functionality. Worked with Object storage with S3.
- Improved performance of **Airflow** by exploring and implementing the most suitable configurations.
- Configured **Spark streaming** to receive real-time data from **Apache Flume** and store the stream data using Scala to Azure Table and **DataLake** is used to store and do all types of processing and analytics. Created data frames using **Spark Dataframes**. Worked with Kudu. Worked with the administration team.
- Designed cloud architecture and implementation plans for hosting complex app workloads on **MS Azure**.
- Performed operations on the transformation layer using **Apache Spark RDD**, **Data frame APIs**, and **Spark SQL** and applied various aggregations provided by Spark framework.
- Provided real-time insights and reports by mining data using **Spark Scala functions**. Optimized existing Scala code and improved the cluster performance. Work experience with Java.
- Documented best practices and guidelines for Cosmos DB configurations, data modeling, and ETL processes, contributing to team knowledge sharing.
- Processed huge datasets by leveraging **Spark Context**, **Spark SQL,** and **Spark Streaming**.
- Enhanced reliability of Spark cluster by continuous monitoring using **Log Analytics** and **Ambari** WEB UI.
- Improved the query performance by transitioning log storage from **Cassandra** to **Azure SQL** Datawarehouse.
- Implemented custom-built input adapters using **Spark, Hive, and Sqoop** to ingest data for analytics from various sources **(Snowflake, MS SQL, MongoDB)** into **HDFS**. Imported data from web servers and Teradata using **Sqoop, Flume, and Spark Streaming API**.
- Improved efficiency of large datasets processing using Scala for concurrency support and parallel processing.
- Developed map-reduce jobs using Scala for compiling program code into bytecode for the **JVM** for data processing. Ensured faster data processing by developing Spark jobs using Scala in a test environment and used Spark SQL for querying. Worked with AKS service.
- Improved processing time and efficiency by using Spark applications like batch interval time, level of parallelism, memory tuning. Monitored workflows for daily incremental loads from **RDBMS (MongoDB, MS SQL, MySQL)**.
- Implemented indexing to data ingestion using **Flume** sink to write directly to indexers deployed on a cluster.
- Delivered data for analytics and Business intelligence needs by managing workloads using **Azure Synapse**.

- Improved security by using **Azure DevOps** and **VSTS (Visual Studio Team Services)** for **CI/CD**, Active Directory, and Apache Ranger for authentication. Managed resources and scheduling across the cluster using **Azure Kubernetes Service.**

Environment: Hadoop, Spark, Hive, Sqoop, HBase, Flume, Ambari, Scala, AKS, MS SQL, Cosmos DB, MySQL, Snowflake, MongoDB, Git, Java, Data Storage Explorer, Python, Azure (Data Storage Explorer, ADF, AKS, Blob Storage)

Client: Charles Schwab Corporation - Westlake, Texas **Oct 2019 – Nov 2021**
Data Engineer
Responsibilities:

- Participated in the analysis, design, and development phase of the Software Development Lifecycle (SDLC).
- Utilized JIRA for project management and GitHub for version control in an agile environment with sprint planning, scrum calls, and retrospectives.
- Designed, developed, and implemented big data solutions using tools such as Kafka, Cassandra, Apache Spark, Spark Streaming, HBase, Flume, Impala, HDFS, MapReduce, Hive, Pig, BDM, Sqoop, Flume, Oozie, and Zookeeper.
- Worked with various programming languages such as SQL, PL/SQL, Python, R, PySpark, Pig, Hive QL, Scala, Shell, Python Scripting, and Regular Expressions.
- Collaborated on Spark components such as RDD, Spark SQL (Data Frames and Dataset), and Spark Streaming.
- Managed cloud infrastructure on AWS, Azure, and GCP for deploying and managing big data solutions. Work experience with Java.
- Conducted performance analysis and optimized AKS clusters for resource efficiency.
- Implemented ETL processes to efficiently move and transform data between Cosmos DB and other data sources, improving data consistency.
- Worked with various databases including Oracle, Teradata, MySQL, SQL Server, and NoSQL Database (HBase, MongoDB) for data extraction and processing. Also, Experience with object storage like S3.
- Developed and maintained ETL pipelines using tools like Sqoop, Flume, and Oozie to extract, transform, and load data into Hadoop.
- Developed and deployed machine learning models for applications such as predictive analytics, recommendation systems, and natural language processing, using Python and leading ML libraries.
- Created data processing pipelines using distributed frameworks such as Apache Spark, Spark Streaming, and HBase.
- Implemented and optimized MapReduce jobs for large dataset processing using Hive and Pig.
- Developed REST APIs using Python with Flask and Django frameworks and integrated various data sources.
- Designed and deployed multi-tier applications using AWS services like EC2, Route53, S3, Databricks, RDS, Dynamo DB, SNS, SQS, IAM, focusing on high-availability and fault tolerance. Worked with Apache ozone. Worked with Minio.
- Collaborated with cross-functional teams to comprehend business objectives and translate them into data and ML solutions.
- Utilized Apache Spark with Python for Big Data Analytics and Machine Learning, including Spark ML and MLlib.
- Supported cloud instances running Linux and Windows on AWS, managing security groups and virtual private clouds.
- Configured Amazon EC2, S3, Elastic Load Balancing, and Security Groups in AWS.
- Automated backups and AMI creation using AWS CLI.
- Installed and configured automation tools like Puppet and Chef.
- Worked on OpenShift and Kubernetes for containerization and orchestration.
- Created Jenkins CI/CD pipelines for building, testing, and delivering artifacts.
- Managed Jenkins artifacts in Nexus repository.
- Set up monitoring with tools like Splunk, AppDynamics, and CloudWatch.
- Configured JIRA as a defect tracking system with custom workflows and plugins.

Environment: AWS, Ansible, ANT, MAVEN, Jenkins, Bamboo, Splunk, Confluence, GCP, Cosmos DB, AKS, Java, Bitbucket, GIT, JIRA, Python, SSH, Shell Scripting, Docker, JSON, Kubernetes, Nagios, Red Hat Enterprise Linux, Terraform, Kibana.

Client: Tesco , Bangalore, India Sept 2017 – Sep 2019
Title: Big Data Engineer
Roles and Responsibilities:

- Worked on developing **ETL** pipelines to transfer legacy systems data to a **Hadoop** cluster. Involved in data cleansing, processing, validation of business requirements, and designing functional specifications for schema and table creations and optimized the query performance using **HIVE**.
- Built, configured, and maintained Hadoop cluster using **Hortonworks** distribution based on business requirements.
- Leveraged Apache bigdata Hadoop components like **HDFS, MapReduce, Yarn, HBase, Sqoop, Pig.**
- Implemented end-to-end **ETL** pipelines using **Python** and **SQL** for high-volume analytics. Reviewed use cases before onboarding to **HDFS**.
- Responsible for loading, managing, and reviewing terabytes of log files using **Ambari** web UI.
- Migrated data from traditional **RDBMS** to **HDFS** using **Sqoop**. Ingested data, from **MS SQL**, **Teradata,** and **Cassandra** databases.Batch processing with Dataproc on spark. Used Strom for large data.
- Worked with the cluster using **Azure Kubernetes Service.**
- Conducted code reviews with Java and implemented best practices to ensure code quality and maintainability.
- Identified required tables, views and exported them to **Hive** and performed ad-hoc queries using **Hive joins**, **Partitioning**, **Bucketing** techniques for faster data access. Worked with Ceph.
- Worked on **Hive** integration with **Spark SQL** scripts for performance enhancement. Implemented file compression techniques using **Spark.** Worked with **kudu.** Experience with Presto query engine.
- Utilized Terraform to define and manage infrastructure as code for AKS deployments.
- Storing Data Files in Google Cloud S3 Buckets daily basis. Using DataProc, Big Query to develop and maintain GCP cloud base solution.
- Automated data flow between various systems using **Nifi**. Designed dataflow models and complicated target tables to obtain relevant metrics from various sources. Worked with the administration. Experience with Java.
- Developed data pipelines from on premises to cloud using **AWS** services like **DMS**, **S3**, **EC2** and **Redshift**.
- Configured and built **AWS Lambda** functions to automate data ingestion and terminate resources not in use.
- Developed **Bash** scripts to get log files from the FTP server and executed **Hive** jobs to parse them.
- Worked on building continuous integration tools using **Jenkins**. Experience with Apache ozone.
- Automated the **Jenkins** pipeline's integration and delivery service using Groovy scripts. Used Jenkins for CI/CD and SVN for version control. Used Druid for big data streaming.
- Spearheaded a data migration project, successfully transitioning data from a legacy system to Cosmos DB, resulting in improved data accessibility and reduced data transfer times.
- Developed reporting dashboards using **PowerBI** by integrating them to **MS SQL Server** as the back-end database.

Environment: Hortonworks 2.0, Hadoop, Hive, HBase, Sqoop, GCP, Pig, Zookeeper, AKS, CosmosDB, Kafka, Nifi, Ambari, Python, SQL, Java, Teradata, MS SQL, Cassandra, PowerBI, Jenkins, SVN.

Client: Quinnox - Pune, India June 2014 – Aug 2017
Data Engineer

Roles and Responsibilities:

- Participated in the analysis, design, and development phase of the **Software Development Lifecycle (SDLC)**.
- Developed test-driven web applications using **Java J2EE**, **Struts 2.0 framework**, **Spring MVC**, Hibernate framework, **JavaScript,** and **SQL Server** database with deployments on **IBM WebSphere**.
- Integrated Cosmos DB with Azure Functions for real-time processing of data events, enhancing overall system responsiveness.
- Designed and developed NSEP, which is an online web application where students can register, find, search, and apply for the jobs available. Utilized **Java J2EE**, **JavaScript**, **SQL**, **HTML**, **CSS,** and **XML** on **Eclipse**. SQL query on presto.
- Designed & developed a web Portal using Struts Framework, J2EE. Developed newsletter as part of process improvement tasks using HTML and CSS to report the weekly activities.Worked with object storage with S3. Work exp. with Druid for analysis.
- Build data pipelines in airflow in GCP for ETL related jobs using different airflow operators.

- Managed resources and scheduling across the cluster using **Azure Kubernetes Service.** Experience with kudu**.**
- Developed front-end, User Interface using **HTML**, **CSS**, **JSP**, **Struts, Angular,** and **NodeJS,** and session validation using **Spring AOP**. Worked with Ceph. Experience with Worked with Dataproc with spark.
- Extensively used Java multi-threading to implement batch Jobs with **JDK** 1.5 features and deployed it on the **JBoss** server. Worked with large flow of data using Strom.
- Collaborated with DevOps teams to implement CI/CD pipelines for Cosmos DB deployments, reducing manual errors and deployment time.
- Implemented monitoring solutions using Azure Monitor, Prometheus, and Grafana for AKS clusters.
- Ensured High availability and load balancing by configuring and Implementing clustering of Oracle on **WebLogic Server 10.3**.
- Improved productivity by developing an automated system health check tool using **UNIX** shell scripts.

Environment: Java/J2EE, Spring, Oracle, Linux, JDBC, Git, HTML, CSS, GCP, CosmosDB, Angular, NodeJS, Postman, A, Servlets, Struts, JSP, WebLogic, PL/SQL, Eclipse